# Apparent Complexity of Quantized Diffusion: Contour Maps, Aliasing and Moiré

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Code and figures: https://github.com/MiguelPerezI/heat-parity-ca

**Abstract**

Colouring the cells of a finite-difference heat simulation by the parity of $\lfloor q\,u \rfloor$ produces black-and-white frames whose apparent complexity rises from nothing, peaks in a lattice of symmetric concentric islands, and dies. We show that these frames are the contour map of the temperature at spacing $1/q$ with alternating fill, that the rise is aliasing, and that the islands are the replicated centres of an undersampled Fresnel zone plate. Three elementary results make this exact: the band count and the total variation are nonincreasing under Neumann heat flow on a box; the phase-averaged perimeter of the sampled pattern equals a sum over grid edges of a triangle wave of the edge jump, $\sum_e \operatorname{dist}(q|\Delta_e u|, 2\mathbb{Z})$, and the discrete total variation is nonincreasing, so every increase of the perimeter is produced by edges whose jump exceeds one quantum; and the block entropy of the pattern has two non-commuting limits, 0 as the grid is refined and maximal as the quantum is refined. The pictures are therefore not a new cellular automaton but a clean example of how complexity measures of a thresholded field report the sampling rather than the dynamics; we close with a remark on why this matters for proposed laws of increasing functional information.

## 1 The observation

The two-dimensional heat equation $u_t = \kappa \Delta u$ was integrated on a $64 \times 64$ grid with an explicit scheme, starting from a hot square ($u = 100$ on a $24 \times 24$ block, 0 elsewhere), and displayed as a height field in which each facet is coloured white or black according to the parity of $\lfloor 10\,u \rfloor$. The frames (Figure 1) begin uniformly black, erupt into a fine noise-like texture around the square, organize into a square-symmetric lattice of concentric islands, and finally fade to a few rings and then to a single colour. The sequence looks like a cellular automaton with a rich transient, and the islands look like eddies. Neither is the case, and the point of this note is to say precisely what they are.

Throughout, $\Omega = (0, L)^n$, $u$ solves $u_t = \kappa \Delta u$ with homogeneous Neumann data, $q > 0$ is the quantization frequency (quantum $1/q$), $\theta \in [0, 1)$ is a phase, and the *pattern* is

$$b_\theta(x, t) = \lfloor q\,u(x, t) + \theta \rfloor \bmod 2. \tag{1}$$

The grid version samples $u$ at the $N^n$ nodes of the explicit scheme with reflected ghost values (the ternary-cube rule of [1] with zero flux), under $\sum_m r_m \le \frac{1}{2}$. We write $E$ for the set of grid edges and $\Delta_e u$ for the jump of $u$ across the edge $e$.

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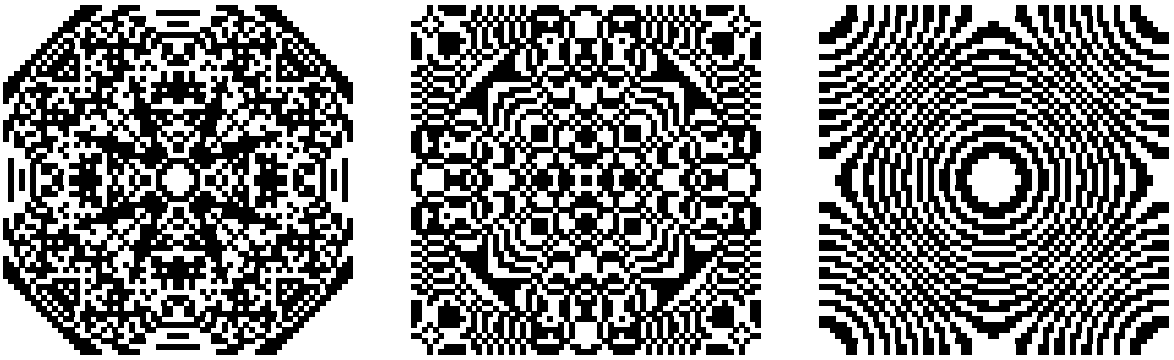


Figure 1: Parity of $\lfloor 10u \rfloor$ on the $64 \times 64$ grid at steps 150, 1000 and 2000 of the explicit scheme with $\Delta t = 2 \cdot 10^{-4}$, $\kappa = 0.2$, reflecting walls. The field has a single maximum throughout.

So (1) is the contour map of $u$ with isotherms at the levels $(k-\theta)/q$ and alternating fill. The pattern is an *observable* of $u$, not a state: two fields with the same pattern evolve to different patterns, so no local rule on black and white cells reproduces the frames, and by the usual definition (a map on configurations over a finite alphabet given by a local rule [7]) this is not a cellular automaton.

## 2 What is monotone in the continuum

**Theorem 2.1.** *Let $u$ solve the Neumann heat equation on the box $\Omega$.*

1. *The band count $B_q(t) = \#\{k : (k-\theta)/q \in [\min u(\cdot,t), \max u(\cdot,t)]\}$ is nonincreasing for every $\theta$.*
2. *The total variation $\mathrm{TV}(t) = \int_\Omega |\nabla u(x,t)|\, dx$ is nonincreasing.*
3. *The phase-averaged total isotherm length equals the total variation exactly:*

$$\int_0^1 \sum_{k\in\mathbb{Z}} \mathcal{H}^{n-1}\big(\{u(\cdot,t) = (k-\theta)/q\}\big)\, d\theta \;=\; q\ \mathrm{TV}(t), \tag{2}$$

*and is therefore nonincreasing.*

*Proof.* (1) is the maximum principle [5]: $\max u$ is nonincreasing and $\min u$ nondecreasing. (2) Reflect $u$ evenly across every face of the box; the result $\tilde{u}$ is a $2L$-periodic solution of the heat equation on the torus $\mathbb{T} = (\mathbb{R}/2L\mathbb{Z})^n$, smooth for $t > 0$. Each component $\partial_m \tilde{u}$ is again a periodic heat solution, so $\nabla\tilde{u}(t) = K_{t-s} * \nabla\tilde{u}(s)$ for $t > s$, where $K$ is the periodic heat kernel, $K \geq 0$, $\int K = 1$. Convexity of the Euclidean norm gives $|\nabla\tilde{u}(t)| \leq K_{t-s} * |\nabla\tilde{u}(s)|$ pointwise; integrating over $\mathbb{T}$ and dividing by $2^n$ gives $\mathrm{TV}(t) \leq \mathrm{TV}(s)$. (3) By the coarea formula, $\int_\Omega |\nabla u| = \int_{\mathbb{R}} \mathcal{H}^{n-1}(\{u = s\})\, ds$; the left side of (2) is $\int_{\mathbb{R}} \mathcal{H}^{n-1}(\{u = s\})\, q\, ds$ after the substitution $s = (k-\theta)/q$, $\theta \in [0,1)$, $k \in \mathbb{Z}$. □

For a fixed phase the sum in (2) is a Riemann sum of the coarea integral, hence within $o(1)$ (as $q \to \infty$) of a nonincreasing function. In the continuum the quantized contour map has no rise of complexity, for any quantum, in the phase-averaged sense.

## 3 The exact perimeter formula on the grid

Let $P_h$ be the number of grid edges whose two cells differ in colour, and $C_h = \sum_{e\in E} |\lfloor qu_a + \theta \rfloor - \lfloor qu_b + \theta \rfloor|$ the number of isotherm crossings, where $a, b$ are the ends of $e$. Let $\mathrm{TV}_h = \sum_{e\in E} |\Delta_e u|$ and let $\varphi(x) = \mathrm{dist}(x, 2\mathbb{Z})$ be the triangle wave of period 2 with $\varphi(x) = x$ on $[0,1]$.

**Theorem 3.1.** *1. (Discrete TVD.) For each axis $m$, $\sum_{\mathbf{i}} |u_{\mathbf{i}+\mathbf{e}_m} - u_{\mathbf{i}}|$ is nonincreasing under the explicit scheme with reflected ghosts and $\sum_m r_m \le \frac{1}{2}$. Hence $\mathrm{TV}_h$ is nonincreasing.*

*2. (Random phase.) With $\theta$ uniform on $[0,1)$,*

$$\mathbb{E}_\theta C_h = q\ \mathrm{TV}_h, \qquad \mathbb{E}_\theta P_h = \sum_{e \in E} \varphi\big(q|\Delta_e u|\big), \tag{3}$$

*both exactly.*

*3. Consequently: if $q|\Delta_e u| \le 1$ on every edge then $\mathbb{E}_\theta P_h = q\,\mathrm{TV}_h$ and is nonincreasing; always $\mathbb{E}_\theta P_h \le \min(q\,\mathrm{TV}_h, |E|)$; every increase of $\mathbb{E}_\theta P_h$ in time is produced by edges whose jump exceeds one quantum and is decreasing through a folded branch of $\varphi$; and $\mathbb{E}_\theta P_h$ is nonincreasing from the first time $t^*$ at which $\max_e q|\Delta_e u| \le 1$.*

*Proof.* (1) On the reflected periodic grid the scheme is a convex combination of translates, $u^{l+1} = \sum_j c_j\, \sigma_j u^l$ with $c_j \ge 0$, $\sum c_j = 1$, by the CFL condition. The axis difference $D_m u = \sigma_{\mathbf{e}_m} u - u$ satisfies the same recursion, and a convex combination of translates is an $\ell^1$-contraction (the linear case of Harten's TVD criterion [6]), so $\|D_m u^{l+1}\|_1 \le \|D_m u^l\|_1$. Restricting the periodic sum to the box changes it by the factor $2^n$. (2) Under a uniform shift the number of levels $(k-\theta)/q$ inside an interval of length $\ell$ is $\lfloor q\ell \rfloor$ with probability $1-\{q\ell\}$ and $\lfloor q\ell \rfloor + 1$ with probability $\{q\ell\}$. Its expectation is $q\ell$, which gives the first identity. The two ends of the edge differ in colour iff that number is odd; if $\lfloor q\ell \rfloor$ is even this has probability $\{q\ell\}$, otherwise $1-\{q\ell\}$, and both equal $\varphi(q\ell)$. (3) follows from $\varphi(x) = x$ on $[0,1]$, $\varphi \le \min(x,1)$, and the fact that $\varphi$ is increasing only on the branches $[2j-1, 2j]$, $j \ge 1$, which require $x > 1$. □

Theorem 3.1 explains the frames. Initially the jumps across the edge of the hot square are a thousand quanta, $\varphi$ is folded a thousand times, and the expected perimeter per such edge is about $\frac{1}{2}$. As the front spreads the number of edges in the folded regime grows, so the perimeter grows, although $q\,\mathrm{TV}_h$ is falling all along. When the largest jump passes below one quantum, the perimeter becomes $q\,\mathrm{TV}_h$ itself and decays with it. Table 1 shows the identity (3) against a Monte-Carlo average over 200 phases on the setup of Figure 1, and the peak at the step where the largest jump crosses one quantum.

| step | $q\,\mathrm{TV}_h$ | $P_h(\theta=0)$ | $\overline{P_h}$ (MC) | $\sum_e \varphi(q\lvert\Delta_e u\rvert)$ | $\max_e q\lvert\Delta_e u\rvert$ |
|---|---|---|---|---|---|
| 20 | 95999.7 | 1136 | 1256.2 | 1257.3 | 155.95 |
| 150 | 87103.5 | 3600 | 3613.3 | 3610.7 | 51.75 |
| 400 | 59437.5 | 4032 | 4004.1 | 4004.7 | 22.19 |
| 1000 | 22542.9 | 4232 | 4285.2 | 4286.4 | 5.75 |
| 1500 | 10006.7 | 3560 | 3552.3 | 3555.3 | 2.22 |
| 2000 | 4435.4 | 4432 | 4436.8 | 4435.4 | 0.92 |
| 2500 | 1964.6 | 1976 | 1963.5 | 1964.6 | 0.39 |
| 3000 | 870.0 | 896 | 867.3 | 870.0 | 0.17 |
| 4000 | 170.5 | 184 | 172.9 | 170.5 | 0.03 |

Table 1: The random-phase identity (3) on the $64 \times 64$ run of Figure 1, $q = 10$. The discrete total variation never increased; the perimeter peaks where the largest jump crosses one quantum; from then on $\overline{P_h} = q\,\mathrm{TV}_h$ to rounding.

## 4 Entropy: the limits do not commute

Let $H_{q,h}(t)$ be the Shannon entropy of the empirical distribution of $2 \times 2$ blocks of $b_\theta$ on the grid of spacing $h$.

**Theorem 4.1.** *Fix $t > 0$.*

1. *For fixed $q$, $\lim_{h\to 0} H_{q,h}(t) = 0$.*
2. *For fixed $h$, if $1, u(x_1,t), \dots, u(x_M,t)$ are linearly independent over $\mathbb{Q}$, where $x_j$ are the grid nodes, then $(q\,u(x_j,t) \bmod 2)_j$ is equidistributed on $(\mathbb{R}/2\mathbb{Z})^M$ as $q \to \infty$, and the Cesàro limit over $q$ of $H_{q,h}(t)$ is the maximum $\log_2 16 = 4$ bits.*

*Proof.* (1) For $t > 0$ the isotherms are smooth curves of finite total length $\Lambda_q(t)$; a $2 \times 2$ block is non-uniform only if an isotherm passes within distance $2h$ of it, so the fraction of non-uniform blocks is $O(h\Lambda_q(t))$, and the entropy of a distribution concentrated on one symbol with mass $1 - O(h)$ is $O(h \log(1/h))$. (2) is Weyl's equidistribution theorem (Kronecker–Weyl) for the sequence $q \mapsto (qu(x_j,t))_j$ on the torus, under the stated rational independence; the uniform distribution on $\{0,1\}^4$ maximizes the block entropy. □

The square-symmetric initial data of the experiment violate the independence in (2), since $u(x_j) = u(\sigma x_j)$ for the symmetries $\sigma$ of the square, and the achievable entropy is reduced by the orbit structure. That is why the aliased frames look like symmetric designs rather than noise. Measured on the frames of Figure 1, the block entropy reaches 3.99 bits at step 400; this is regime (2), a statement about the sampling, not about the heat equation.

## 5 The islands are moiré

Where $u$ is locally linear, $\lfloor q(a \cdot x) + \theta \rfloor \bmod 2$ on the lattice is a two-colour digital plane, whose one-dimensional sections are Sturmian words; these are the striped regions. Where $u$ is locally quadratic, near the maximum and in the tails, the pattern is a sampled Fresnel zone plate. An undersampled zone plate replicates its centre [4, 9]: neighbouring cells agree in colour wherever $q\,h\,\partial_m u$ is near an even integer, so fake centres appear along the curves $\{qh\,\partial_m u \in 2\mathbb{Z}\}$, $m = 1, 2$. Figure 2 shows the effect in its purest form: the state at $t = 0.03$, when the field still carries 832 isotherms and the gradient at the edge of the square is about forty quanta per cell even on a $256^2$ grid. The true contour map would be a set of tightly nested rounded squares; what the lattice sees instead is a central zone plate surrounded by a ring of replica centres, each a small zone plate of its own, arranged on the curves where $qh\,\partial_m u$ passes through even integers, with the quasi-periodic stripes of digital planes between them. Every feature of this picture is a property of the sampling: the field has one maximum, and its isotherms never cross. Figure 3 shows the later state $t = 0.4$ at 64 and 256 cells with the same quantum: the island lattice is present at 64 and absent at 256, where the frame is an ordinary contour map. Table 2 gives the measures. With the quantum scaled as $1/N$ instead, the measures are constant in $N$ (perimeter per cell 0.879, 0.894, 0.893): the texture depends on the single ratio $qh|\nabla u|$ and is scale-free undersampling.

## 6 What is not claimed, and what the lesson is

The number of components of the pattern, and the number of local maxima of $u$, are not monotone in general in two dimensions [3]; nor is the entropy of the value histogram of $u$. There are no eddies: for $t > 0$ the field is analytic with smooth nested isotherms. The lesson is methodological. Perimeter, block entropy and component counts of a thresholded

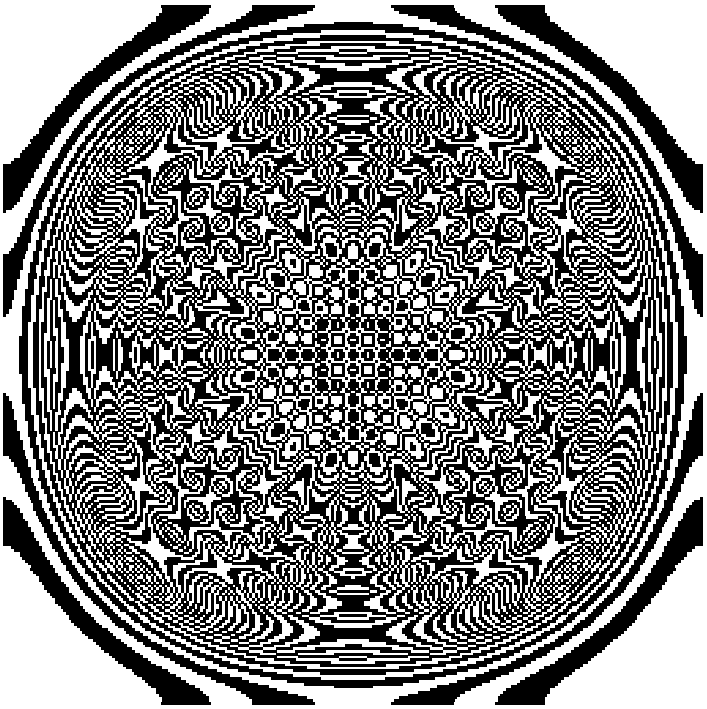


Figure 2: The parity pattern at $t = 0.03$ on a $256^2$ grid, $q = 10$, 832 isotherms. An undersampled Fresnel zone plate: the central rings are the real contour map of the peak, the surrounding lattice of small concentric centres is moiré, replicas created where the local fringe frequency matches a multiple of the sampling frequency, and the striped sectors between them are digital planes. The figure is vector graphics; zooming in shows the individual cells.

| $N$ | bands | perimeter/$N^2$ | $H_{2\times2}$ (bits) | components |
|---|---|---|---|---|
| 64 | 34 | 1.082 | 3.93 | 905 |
| 128 | 36 | 0.564 | 3.45 | 88 |
| 256 | 37 | 0.287 | 2.61 | 92 |

Table 2: Resolution test at $t = 0.4$, $q = 10$ fixed. Perimeter per cell halves with each doubling, as a fixed set of smooth curves does; the islands collapse into the real rings.

simulation measure the sampling as much as the dynamics; the phase-averaging identity (3) is a diagnostic, since a phase-averaged perimeter exceeding $q\,\mathrm{TV}_h$ is aliasing by Theorem 3.1(3). Real complexity requires the nonlinearity inside the dynamics: putting the modulus into the update rule instead of the display gives Wolfram's continuous cellular automata [2, ch. 4], coupled map lattices in the sense of [8], which do have localized structures; their interaction with reflecting walls is the subject of a companion paper [13].

**A remark on functional information.** Wong et al. [10] have proposed a law of increasing functional information: the functional information of a system increases when many configurations of the system undergo selection for one or more functions. Functional information, in the sense of Szostak and Hazen [11, 12], is $I(E_x) = -\log_2 F(E_x)$, where $F(E_x)$ is the fraction of all configurations that perform a given function to degree at least $E_x$; it is defined only relative to a function, and the proposed law makes selection, not mere dynamics, the source of its growth. Quantized diffusion is a clean example of why the restriction to a function is necessary. The quantized heat frames have no function and undergo no selection; every



Figure 3: Left: $64 \times 64$ grid at $t = 0.4$ ($q = 10$). Right: the same state on a $256 \times 256$ grid with the same quantum. The islands of the left frame are replicated zone-plate centres; the right frame is the contour map of the field, with its mirror images in the four corners produced by the reflecting walls.

quantity intrinsic to the field, the band count, the total variation, the gradient, decreases monotonically (Theorems 2.1 and 3.1); and yet the function-free complexity proxies that are commonly read as evidence of "ratcheting" complexity, perimeter, block entropy, component count, rise from zero to near their maxima before they fall. The rise is produced entirely by the observer's choices of quantum and grid, and Theorem 4.1 shows that these choices can make the block entropy anything between zero and its maximum. The random-phase identity (3) is the diagnostic that separates the two cases: a phase-averaged perimeter above $q\,\mathrm{TV}_h$ is sampling, not structure. A system has to do more than produce pictures that look complex before the question of selection can even be asked; in the companion paper [13] we take automata that do sustain persistent localized structures and ask whether the walls of a box act on them as a selecting environment.

## Code

All scripts, the figures and the source of this article are public at `https://github.com/MiguelPerezI/heat-parity-ca`: `analyze.py` reproduces the frames and the measures of Table 2, `restest.py` the resolution test, and `phase_theorem.py` checks Theorem 3.1 and produces Table 1.

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